# Lecture 16

## Variable Coefficient Equations and Reduction of Order

We have so far taken for granted that the constant-coefficient equation

$$ay'' + by' + cy = f(t)$$

has solutions that are valid for some value(s) of $t$. In fact, they are valid where the function $f(t)$ is continuous.

If we consider the variable coefficient case for a linear 2nd-order equation

$$a(t)y'' + b(t)y' + c(t)y = f(t)$$

we may expect that any solutions we find are only valid where all four functions $a(t)$, $b(t)$, $c(t)$, and $f(t)$ are continuous simultaneously.

Recall: $y' + P(x)y = Q(x)$ P, Q cont.

**Theorem** (Existence and Uniqueness of Linear 2nd-Order Equations)**:** Consider the linear 2nd-order equation

$$y'' + p(t)y' + q(t)y = g(t) \qquad \text{standard form}$$

and suppose that $p(t)$, $q(t)$, and $g(t)$ are continuous on some interval $I$ containing $t_0$. Then for any choice of initial conditions $y(t_0) = y_0$, $y'(t_0) = y_1$, there is a unique solution $y(t)$ on $I$ to the IVP

$$y'' + p(t)y' + q(t)y = g(t), \quad y(t_0) = y_0, \quad y'(t_0)y_1$$

**Example:** On what interval is a unique solution guaranteed by the E-U Theorem for the following equation at $t = 1$?

$a(t)$

$$(t-3)y'' + y' + \sqrt{t}y = \ln(t)$$

write in standard form: $y'' + \frac{1}{t-3}y' + \frac{\sqrt{t}}{t-3}y = \frac{\ln(t)}{t-3}$
(divide by $t-3$)

$p(t) = \frac{1}{t-3}$ : cont. $t \neq 3$, $(-\infty, 3) \cup (3, \infty)$

$q(t) = \frac{\sqrt{t}}{t-3}$ : cont. $t \neq 3, t \geq 0$, $[0,3) \cup (3, \infty)$

$g(t) = \frac{\ln(t)}{t-3}$ : cont. $t \neq 3, t > 0$ $(0,3) \cup (3, \infty)$

$I_1$ $I_2$

· All of $p, q,$ and $g$ are continuous on $(0,3) \cup (3, \infty)$

Want a solution around $t = 1$, it is guaranteed by E-U Theorem to be valid on $(0,3)$.

Once again, this E-U theorem does not tell us how to find a solution, but only that such a solution exists. In fact, unlike the 1st-order case, there is no general method to solve a linear 2nd-order equation with variable coefficients (the same holds true for higher-order equations as well). So we must rely on various solving techniques to find solutions to variable coefficient equations.

We will start our discussion of solving variable coefficient equations by defining a class of such equations that we can solve in general.

**Definition:** A linear equation of the form $at^2y'' + bty' + cy = f(t)$

ind

$$a_n t^n y^{(n)} + a_{n-1} t^{n-1} y^{(n-1)} + \cdots + a_1 t y' + a_0 y = f(t)$$

where $a_0, a_1, \ldots, a_n$ are constants, is known as a **Cauchy-Euler equation** (or **equidimensional** equation).

**Example:** Are the following equations Cauchy-Euler (C-E) equations?

- $3t^2y'' + 11ty' - 3y = \sin(t)$ linear

  Yes, C-E

- $2y'' - 3ty' + 11y = 3t - 1$ linear

  multiply by $t^2 \Rightarrow 2t^2y'' - 3t^3y' + 11t^2y = 3t^3 - t^2$

  not C-E

- $4t^2y''' - 5y' = t^3$

  multiply by $t \Rightarrow 4t^3y''' - 5ty' = t^4$ do not need all derivatives

  Yes, C-E

We will approach solving a 2nd-order C-E equation in the same way as we did for solving the constant-coefficient 2nd-order case. First, we will consider a homogeneous equation:

$$at^2y'' + bty' + cy = 0, \quad t > 0 \tag{16.1}$$

Note: By the E-U Theorem we are only guaranteed solutions for the intervals $t < 0$ or $t > 0$ (we usually restrict to $t > 0$) as a C-E equation in standard form is

$$y'' + \frac{b}{at}y' + \frac{c}{at^2}y = 0$$

$p(t) = \frac{b}{at}, \quad q(t) = \frac{c}{at^2}, \quad t \neq 0$

and the functions $p(t) = \frac{b}{at}$ and $q(t) = \frac{c}{at^2}$ are discontinuous at $t = 0$.

$y' = rt^{r-1}$

$at^2y'' = at^2(r(r-1)t^{r-2}) = a(r(r-1))t^r$

To solve the homogeneous C-E equation we will try a solution of the form $y = t^r$ as the product of $t^r$ with $y^{(i)}$ guarantees that each term will be a constant times $t^r$ allowing us to condense one side of the equation down to a form with only a single term.

We have

$$y = t^r, \quad y' = rt^{r-1}, \quad y'' = r(r-1)t^{r-2}$$

and substituting into (16.1) we have

$$\begin{aligned} at^2y'' + bty' + cy &= at^2(r(r-1)t^{r-2}) + bt(rt^{r-1}) + ct^r \\ &= a(\overbrace{r(r-1)}^{=2-r})t^r + brt^r + ct^r \\ &= t^r[ar^2 - ar + br + c] \\ &= t^r[\underbrace{ar^2 + (b-a)r + c}_{=0}], \quad t > 0 \\ &\overset{\text{want!}}{=} 0 \end{aligned}$$

and if we want this to solve the equation we must have $ar^2 + (b-a)r + c = 0$ as $t \neq 0$.

The equation $ar^2 + (b-a)r + c = 0$ is the **characteristic equation** (or

solution $r = \dfrac{-(b-a) \pm \sqrt{(b-a)^2 - 4ac}}{2a}$



**auxiliary equation**) for the 2nd-order Cauchy-Euler equation.

This is a quadratic polynomial in $r$ and just as in the constant-coefficient case, we have three separate cases to consider as solution types.

For the 2nd-order homogeneous C-E equation

$$at^2y'' + bty' + cy = 0, \quad t > 0$$

$= r_2 t^{r_1+r_2-1} - r_1 t^{r_2+r_1-1}$
$= t^{r_1+r_2-1}(r_2 - r_1) \neq 0.$
$t>0$ $t>0$

the associated characteristic equation is $ar^2 + (b-a)r + c = 0$ and we have the following cases:

$t^{r_1}, t^{r_2}$ LI for $t>0$ $\begin{vmatrix} t^{r_1} & t^{r_2} \\ r_1 t^{r_1-1} & r_2 t^{r_2-1} \end{vmatrix}$

**Case 1:** Two distinct real roots $r_1 \neq r_2$ (i.e. $(b-a)^2 - 4ac > 0$)

analogous to $c_1 e^{r_1 t} + c_2 e^{r_2 t}$

In this case, the general solution is $y_h(t) = c_1 t^{r_1} + c_2 t^{r_2}$

$e^{rt}$ $te^{rt}$

**Case 2:** One repeated real root $r$ (i.e. $(b-a)^2 - 4ac = 0$)

In this case, the general solution is $y_h(t) = c_1 t^r + c_2 t^r \ln(t)$.

Note: In the constant coefficient case if we had a repeated root to the characteristic equation we would use the homogeneous solution $y_h = c_1 e^{rt} + c_2 t e^{rt}$ by multiplying the $e^{rt}$ by $t$ to get a second linearly independent solution. We do a similar multiplication here but instead multiply by $\ln(t)$. We will see exactly why these multiplications arise when discussing Reduction of Order.

**Case 3:** Two complex conjugate roots $r = \alpha + i\beta$, $\beta \neq 0$ (i.e. $(b-a)^2 - 4ac < 0$)

In this case, we have $t^{\alpha+i\beta} = t^{\alpha}t^{i\beta}$. We can change the base by using logarithms to say $t = e^{\ln(t)}$ and with Euler's formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ we have

$$t^{\alpha+i\beta} = t^{\alpha}t^{i\beta} = t^{\alpha} \cdot (e^{\ln(t)})^{i\beta} = t^{\alpha}e^{i(\beta\ln(t))} = t^{\alpha}[\cos(\beta\ln(t)) + i\sin(\beta\ln(t))].$$

By the same argument as in the constant coefficient case we get two linearly independent real solutions by taking the real and imaginary parts. This gives us the solution

$$y_h(t) = c_1 t^{\alpha}\cos(\beta\ln(t)) + c_2 t^{\alpha}\sin(\beta\ln(t)).$$

**Example:** Find the homogeneous solution to

$$t^2y'' + 3ty' - 3y = 0, \quad t > 0$$

yes C-E!

char eqn: $ar^2 + (b-a)r + c = 0$, $a=1$, $b=3$, $c=-3$

$\Rightarrow 1 \cdot r^2 + (3-1)r - 3 = 0 \Rightarrow r^2 + 2r - 3 = 0$

$(r+3)(r-1) = 0$

$r = 1, -3$

$y_h = c_1 t^1 + c_2 t^{-3}$

$y = t$, $y' = 1$, $y'' = 0$

$t^2(0) + 3t(1) - 3t =$

$0 + 3t - 3t = 0$ ✓

**Example:** Find the homogeneous solution to

$$t^2y'' + 3ty' + y = 0, \quad t > 0$$

yes C-E!

$a = 1$, $b = 3$, $c = 1$

char eqn: $\underset{a}{1} \cdot r^2 + \underset{b-a}{(3-1)}r + \underset{c}{1} = 0 \Rightarrow r^2 + 2r + 1 = 0$

$(r+1)^2 = 0$

$r = -1, -1$ repeated

$y_h = c_1 t^{-1} + c_2 t^{-1}\ln(t)$

To solve a non-homogeneous C-E equation

$$at^2y'' + bty' + cy = f(t)$$

we must use variation of parameters to find a particular solution (there is no analogous method for undetermined coefficients in the variable coefficient case).

**Example:** Find the general solution to

$$t^2y'' - 3ty' + 3y = 2t^4e^t, \quad t > 0$$

char eqn: $a=1,\ b=-3,\ c=3$

$ar^2+(b-a)r+c=0 \Rightarrow 1\cdot r^2+(-3-1)r+3=0$

$\Rightarrow r^2-4r+3=0 \Rightarrow (r-3)(r-1)=0$

$\Rightarrow r=1,3$ so, $y_h = c_1\underbrace{t}_{y_1} + c_2\underbrace{t^3}_{y_2}$

For variation of parameters, put eqn in standard form!

divide by $t^2$: $\underset{a=1}{y''} - \frac{3}{t}y' + \frac{3}{t^2}y = \underbrace{2t^2e^t}_{f(t)}$

we have $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t & t^3 \\ 1 & 3t^2 \end{vmatrix} = 3t^3 - t^3 = 2t^3$

$W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix} = \begin{vmatrix} 0 & t^3 \\ 2t^2e^t & 3t^2 \end{vmatrix} = 0 - 2t^5e^t$, $W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix} = \begin{vmatrix} t & 0 \\ 1 & 2t^2e^t \end{vmatrix} = 2t^3e^t$

Thus $v_1 = \int \frac{W_1}{W}dt = \int \frac{-2t^5e^t}{2t^3}dt = \int -t^2e^t dt$

$= -t^2e^t + 2te^t - 2e^t$

IBP
$-t^2 \xrightarrow{+} e^t$
$-2t \xrightarrow{-} e^t$
$-2 \xrightarrow{+} e^t$
$0 \quad e^t$

$v_2 = \int \frac{W_2}{W}dt = \int \frac{2t^3e^t}{2t^3}dt = \int e^t dt = e^t$

So $\boxed{y_g = y_h + \underbrace{y_p}_{v_1y_1+v_2y_2} = (c_1t + c_2t^3) + ((-t^2e^t + 2te^t - 2e^t)t + e^t t^3)}$

In general for a variable coefficient linear 2nd-order equation, we have no general method to find a solution. If we can determine the homogeneous solution in some way, then we can use variation of parameters to find any particular solution and thus a full general solution. It can be difficult to determine both linearly independent solutions that make up the homogeneous solution.

Luckily, we only need find one function that satisfies the homogeneous equation and we can construct a linearly independent second solution with the following result:

**Theorem** (Reduction of Order): Let $y_1(t)$ be a solution to the equation

$$y'' + p(t)y' + q(t)y = 0$$

in an interval $I$ with $y_1(t)$ not the zero function ($y_1(t) \not\equiv 0$). Then

$$\boxed{y_2 = y_1 \underbrace{\int \frac{e^{-\int p(t)dt}}{y_1^2(t)}\, dt}_{v}}$$

is a second linearly independent solution.

**Proof:** Consider $y'' + p(t)y' + q(t)y = 0$ $*$

Let $y_2 = v y_1$, $y_2' = v'y_1 + v y_1'$

$y_2'' = v''y_1 + v'y_1' + v'y_1' + v y_1'' = v''y_1 + 2v'y_1' + v y_1''$

Substitute:

$$(v''y_1 + 2v'y_1' + v y_1'') + p(t)(v'y_1 + v y_1') + q(t)[v y_1] = 0 \quad \text{want}$$

Rewrite as 2nd order for $v$.

$$y_1 v'' + (2y_1' + p(t)y_1)v' + \underbrace{(y_1'' + p(t)y_1' + q(t)y_1)}_{*\ =0\ y_1 \text{ is a soln}} v = 0$$

$$\Rightarrow y_1 v'' + (2y_1' + p(t)y_1)v' = 0$$

Let $w = v' \Rightarrow y_1 w' + (2y_1' + p(t)y_1)w = 0$ (1st order)

$$\Rightarrow y_1 \frac{dw}{dt} + (2y_1' + p(t)y_1)\,w = 0 \quad \text{Separable!}$$



$$\Rightarrow y_1 \frac{dw}{dt} = -(2y_1' + p(t)y_1)\,w$$

$$\Rightarrow \int \frac{1}{w}\,dw = -\frac{(2y_1' + p(t)y_1)}{y_1}\,dt = \int\left(-2\frac{y_1'}{y_1} - p(t)\right)dt$$

no constant!

$$\Rightarrow \ln|w| = -2\ln|y_1| - \int p(t)\,dt$$

$$\Rightarrow w = e^{-2\ln|y_1|}\,e^{-\int p(t)dt} \Rightarrow w = y_1^{-2}e^{-\int p(t)dt}$$

$$w = v' \Rightarrow v' = \frac{e^{-\int p(t)dt}}{y_1^2} \Rightarrow \boxed{v = \int \frac{e^{-\int p(t)dt}}{y_1^2}\,dt}$$

**Example:** Suppose that the constant coefficient equation $ay'' + by' + cy = 0$ is such that its associated characteristic equation $ar^2 + br + c = 0$ has a repeated root $r = -\frac{b}{2a}$. (i.e. $b^2 - 4ac = 0$). We then know that $y_1 = e^{-\frac{b}{2a}t}$ is a solution. Find a second linearly independent solution using reduction of order.

$$y_1 = e^{-\frac{b}{2a}t}, \quad ay'' + by' + cy = 0$$

$$\Rightarrow y'' + \frac{b}{a}y' + \frac{c}{a}y = 0 \Rightarrow p(t) = \frac{b}{a}$$

Reduction of Order: $$y_2 = y_1 \int \frac{e^{-\int p(t)dt}}{y_1^2}\,dt = e^{-\frac{b}{2a}t}\int \frac{e^{-\int \frac{b}{a}dt}}{\left(e^{-\frac{b}{2a}t}\right)^2}\,dt$$

$$= e^{-\frac{b}{2a}t}\int \frac{\cancel{e^{-\frac{b}{a}t}}}{\cancel{e^{-\frac{b}{a}t}}}\,dt = e^{-\frac{b}{2a}t}\int 1\,dt$$

$$= e^{-\frac{b}{2a}t}\cdot t$$

So $$y_h = c_1 e^{-\frac{b}{2a}t} + c_2\, t e^{-\frac{b}{2a}t}$$

**Example:** Suppose that the Cauchy-Euler equation $at^2y'' + bty' + cy = 0$ is such that its associated characteristic equation $ar^2 + (b-a)r + c = 0$ has a repeated root $r = -\frac{b-a}{2a}$, (i.e. $(b-a)^2 - 4ac = 0$). We then know that $y_1 = t^{-\frac{b-a}{2a}}$ is a solution. Find a second linearly independent solution using reduction of order.

$$r = \frac{-(b-a) \pm \sqrt{(b-a)^2 - 4ac}}{2a} = \frac{-(b-a)}{2a} \text{ if } (b-a)^2 - 4ac = 0$$

$$y_1 = t^{-\frac{(b-a)}{2a}}, \text{ in standard form } y'' + \frac{b}{at}y' + \frac{c}{at^2}y = 0$$

$$\Rightarrow p(t) = \frac{b}{at}$$

Reduction of Order: $$y_2 = y_1 \int \frac{e^{-\int p(t)dt}}{y_1^2}\,dt$$

$$\Rightarrow y_2 = t^{-\frac{(b-a)}{2a}} \int \frac{e^{-\int \frac{b}{at}dt}}{\left(t^{-\frac{(b-a)}{2a}}\right)^2}\,dt$$

$$= t^{-\frac{(b-a)}{2a}} \int \frac{e^{-\frac{b}{a}\ln(t)}}{t^{-\frac{(b-a)}{a}}}\,dt = t^{-\frac{(b-a)}{2a}} \int \frac{t^{-\cancel{\frac{b}{a}}}}{t^{-\frac{b+a}{a}}}\,dt \quad \} \to \cancel{-\frac{b}{a}} + 1$$

$$= t^{-\frac{(b-a)}{2a}} \int \frac{1}{t}\,dt = t^{-\frac{(b-a)}{2a}} \ln(t)$$

So $$y_h = c_1 t^{-\frac{(b-a)}{2a}} + c_2 t^{-\frac{(b-a)}{2a}} \ln(t)$$

**Example:** Find a general solution to the variable coefficient equation

$$ty'' - (t+1)y' + y = t^2e^{2t} \quad (= (te^t)^2), \quad t > 0$$

Assume a solution is of the form $y = Ce^t$ for the homogeneous eqn:

$y' = Ce^t$, $y'' = Ce^t$

sub:

$$t(Ce^t) - (t+1)Ce^t + Ce^t \overset{\text{homogeneous}}{=} 0$$

$$\Rightarrow \underbrace{tCe^t - tCe^t}_{0} \underbrace{- Ce^t + Ce^t}_{0} \overset{?}{=} 0 \checkmark$$

Use $y_1 = e^t$, then $y_2 = y_1 \int \frac{e^{-\int p(t)dt}}{y_1^2} dt =$

Standard form: $\boxed{y'' - \frac{(t+1)}{t}y' + \frac{1}{t}y = te^{2t}}$, $p(t) = -1 - \frac{1}{t}$

So $y_2 = e^t \int \frac{e^{-\int -1 - \frac{1}{t} dt}}{e^{2t}} dt = e^t \int \frac{e^{+t+\ln(t)}}{e^{2t}} dt = e^t \int \frac{te^t}{e^{2t}} dt$

$$= e^t \int te^{-t} dt \overset{IBP}{=} e^t(-te^{-t} - e^{-t}) = -t-1$$

$t \quad e^{-t}$
$1 \quad -e^{-t}$
$0 \quad e^{-t}$

var. Par.: $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^t & -t-1 \\ e^t & -1 \end{vmatrix} = -e^t - e^t(-t-1)$
$= te^t$

$W_1 = \begin{vmatrix} 0 & y_2 \\ f(a) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & -t-1 \\ te^{2t} & -1 \end{vmatrix} = 0 - t(-t-1)e^{2t} = t(t+1)e^{2t}$

$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(a) \end{vmatrix} = \begin{vmatrix} e^t & 0 \\ e^t & te^{2t} \end{vmatrix} = te^{3t}$

$v_1 = \int \frac{W_1}{W} dt = \int \frac{t(t+1)e^{2t}}{te^t} dt$
$= \int (t+1)e^t dt \overset{IBP}{=} te^t$

$v_2 = \int \frac{W_2}{W} dt = \int \frac{te^{3t}}{te^t} dt = \int e^{2t} dt$
$= \frac{1}{2}e^{2t}$

So $\boxed{y_g = y_h + \underbrace{y_p}_{v_1y_1 + v_2y_2} = (c_1e^t + c_2(-t-1)) + (te^t \cdot e^t + \frac{1}{2}e^{2t}(-t-1))}$